



MSA

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Thermodynamic Performance Assessment of a Photovoltaic System

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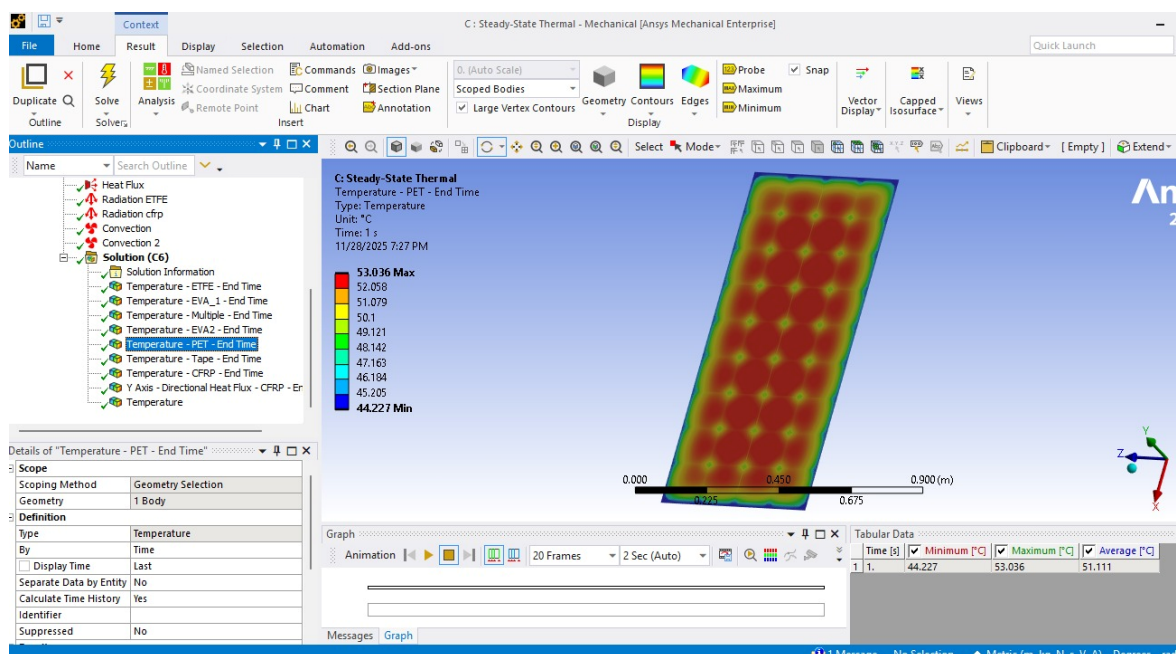
Abstract

A comprehensive thermodynamic and operational performance assessment of a rooftop photovoltaic (PV) system. The study will quantify electrical output, thermal behavior, and energy conversion efficiency under real environmental conditions. Measurements will include irradiance, module temperature, ambient temperature, electrical performance parameters, and inlet/outlet temperatures where thermal analysis is applicable. Using these data, the study will calculate electrical and thermal efficiencies, perform a complete energy-balance analysis, and examine performance variations over time. The outcomes will support optimization of PV system design, operation, and maintenance practices.

1. Background

Photovoltaic systems convert solar radiation directly into electrical energy based on semiconductor physics and PV characteristics as described in PV Systems coursework materials. Their output is strongly influenced by **irradiance**, **temperature**, **tilt angle**, and **system configuration**, all of which affect parameters such as open-circuit voltage, current generation, and maximum power point.

Field performance assessments such as the evaluation of a **3-kW rooftop grid** connected system have demonstrated that real-world conditions, thermal accumulation, aging, dust deposition, and structural deviations cause measurable degradation and efficiency losses



2. Project Objectives

2.1 Primary Objectives

- Perform a complete thermodynamic assessment through measured irradiance, temperatures, and electrical output.
- Calculate electrical efficiency, thermal efficiency, and instantaneous power.
- Develop an energy-balance model including solar input, electrical conversion, and thermal losses (convective, radiative, and conductive).
- Evaluate performance under varying climatic conditions and assess degradation patterns.

2.2 Secondary Objectives

- Estimate greenhouse gas mitigation using measured energy production.
- Provide system-level optimization recommendations related to tilt angle, cooling behavior, shading, cleanliness, and structural stability.
- Compute operational indicators such as capacity utilization hours (CUH).

3. Scope of Work

3.1 System Description

The system consists of rooftop PV modules installed in fixed-tilt configuration.

Module specifications, array configuration, inverter characteristics, and mounting structure will be documented. Measurements of **tilt**, **azimuth**, **shading**, and **surface conditions** will be included to understand their impact on performance.

3.2 Parameters to Be Measured

- **Solar irradiance (W/m^2).**
- **Ambient and module temperatures.**
- **Electrical output: voltage, current, power.**
- Wind speed and environmental conditions.
- Inlet/outlet temperatures and flow rate (if thermal evaluation applies).

4. Methodology

4.1 Experimental Method

Instrumentation includes pyranometers/reference cells, thermocouples, temperature sensors, flow meters, and inverter monitoring tools. Data will be logged at 15–60 second intervals over 1–2 weeks to ensure representative operating conditions.

4.2 Calculations and Thermodynamic Modeling

Electrical power:

$$P_{el} = V \times I \text{ Electrical}$$

Efficiency:

$$\eta_{el} = \frac{P_{el}}{G \times A \text{ Thermal}}$$

Power (if applicable):

$$Q_{th} = \dot{m} C_p (T_{out} - T_{in})$$

Energy balance:

$$Q_{in} = G \times A = P_{el} + Q_{losses}$$

Degradation will be evaluated using corrected performance ratios and daily yield trends. CUH and CO₂ mitigation will be calculated from measured energy production.

5. Background Information

5.1. Solar Angles of The Sun

The Total solar radiation falling on a PV titled surface G_t , depends on the incidence angle, which depends on 8 angles of the sun:

1. **Latitude Angle (L):** The angle measured from the center to the equator.
2. **Declination Angle:** The angle between the sun-earth centerline and the projection of on the equatorial plane.
3. **Hour Angle:** The solar time but in degrees.

$$h = (AST - 12)15$$

$$AST = LST + ET \pm 4(SL - LL) - DS \quad (2.3)$$

where

LST = local standard time.

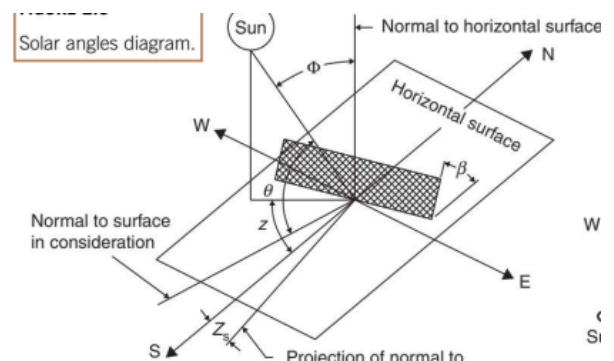
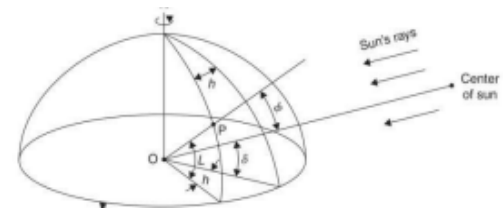
ET = equation of time.

SL = standard longitude.

LL = local longitude.

DS = daylight saving (it is either 0 or 60 min).

4. **Altitude Angle:** Angle between the sun ray's and the horizontal plane.
5. **Zenith Angle:** Angle between the sun ray's and the vertical plane.
6. **Solar Azimuth angle:** Angle of the sun rays measured in the horizontal plane from true south.
7. **Tilted Angle:** Angle between the PV surface and horizontal surface.
8. **Surface Azimuth Angle:** Angle in the horizontal plane, between the line due south and the horizontal projection of the normal to the inclined plane surface.



6. PV design

6.1. Assumptions

There is a total of 26 assumptions that are needed in order to be able to design a suitable PV system, as there are numerous factors that are affected by the day of the year, Temperature, the reflection, absorb, and transparency of the materials of PV as shown in **Figure 6.1**.

Assumptions

- $T_{initial} = 25^{\circ}\text{C}$, - Location Egypt [26.806°N , 30.8025°E]
- $\rho = 0.2$, - Day July 17 $\rightarrow N = 198$
- $h(\text{Solar Noon}) = \text{Zero}$, - $G_{sc} = 1366 \text{ W/m}^2$
- $\bar{H} = 29.7 \text{ MJ/m}^2$, - $\beta_{\text{tilt angle}} = 30^{\circ}$
- **Power_{Req} = 2.9 Kwatt = 2900 watt**
- $e = 1.602 \times 10^{-19}$, - $h = 1.381 \times 10^{-23}$, -
- Sat Current $= 1.7 \times 10^{-8} \text{ A/m}^2$, - S.C Current $= 250 \text{ A/m}^2$
- ETFE Reflection $\rightarrow 0.07$, absorb $= 0.1$, Transparency $= 0.83$
- EVA Reflection $\rightarrow \text{Zero}$, absorb $= 0$, Transparency $= 1$
- Silicon absorb $= 0.97$
- Average Egyptian house (Refrigerator - lighting - washing machine - TV - Fans) $\sim 12 \text{ Kwh/Day} \sim \text{Kwh Cost } 1.95 \text{ L.E}$
 $\sim 2.9 \text{ Kwatt Farm on } 24 \text{ m}^2 \text{ Costing } 90 \text{ K L.E}$

Appliance Power

AC = 1.2 Kw

Refrigerator = 1.8 Kw

Lights = 0.01 Kw

washing machine = 0.8 Kw

Water Heater = 2 Kw

General load (Laptop, Phone, etc) = 1.5 Kwh/Day

Absorption, Transmission and Reflection of Surfaces:

Material	Absorptivity	Transmissivity	Reflectivity
ETFE	0.1	0.83	0.07
EVA	0	1	0
Si	0.97	0	0.03

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Thermal Behavior of Monocrystalline Silicon Solar Cells: A Numerical and Experimental Investigation on the Module Encapsulation Materials

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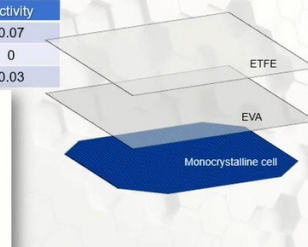


Figure 4.1: Inside of PV

6.2. Extraterrestrial Solar Radiation

Extraterrestrial Solar Radiation is the amount of energy per unit time, at the mean distance of the earth from the sun and the normal surface to sun outside the **atmosphere**, also called **G**. In order to determine the value of **G**, we must calculate some angles of the sun.

□ Solar altitude angle calculation

- * $B = (N - 81) \frac{360}{364}$
- * $ET = 9.87 \sin(2B) - 7.53 \cos(B) - 1.5(B)$
- * $AST = LST + ET \pm 4 (SL - LL) - DS$
- * $h = (AST - 12) / 15$

$B = (198 - 81) \frac{360}{364} = 115.714$ $ST = 12:00 - 178 \text{ min} = 8:38 \text{ Am}$
 $ET = -178 \text{ min},$ $h_{ss} = 99.5, h = 0 \rightarrow \text{Solar noon}$

$\sin \alpha = \cos(\delta) \cos(L) \cos(h) + \sin(\delta) \sin(L)$
 $= \cos(21.21) \cos(26.8206) \cos(0) + \sin(21.21) \sin(26.8206)$
 $= 0.995 \quad \therefore \alpha = 84.38^\circ$

Solar Azimuth Angle

$\sin Z = \frac{\cos(\delta) \sin(h)}{\cos(\alpha)} = 0$

$\text{airmass}(m) = \frac{1}{\sin(\alpha) + 0.5072(6.07995 + \alpha)^{-1.6364}}$
 $= 1.0045$

From table:

$\tau_b = 0.451, \tau_d = 1.976$ $N = 30.13$
 $E = 27.22$

Parameter of airmass:

$b = 1.219 - 0.043\tau_b - 0.151\tau_d - 0.204\tau_b\tau_d = 0.7194$
 $d = 0.202 + 0.852\tau_b - 0.007\tau_d - 0.357\tau_b\tau_d = 0.25427$

* $G_{sc} = 1346 \text{ W/m}^2$ Value = $1 + 0.034 \cos\left(\frac{360 \cdot 194}{365.25}\right)$
 $= 1412.17 \text{ Watt/m}^2$
 $\rightarrow \text{The Extraterrestrial Solar radiation.}$

6.3 Total Solar Radiation on tilted collected surface

Solar Radiation on an arbitrary **tilted** surface having tilt angle **Beta** from the horizontal, and an azimuth angle z .

[2] Direct Solar Radiation:

$$G_{b,n} = G e^{-\tau_{b,n}} = 1412.427 e^{-0.451 \times 1.0046^{0.7154}} \\ = 898.391 \text{ W/m}^2$$

The diffuse Radiation on Horizontal

$$G_{d,h} = G e^{-\tau_{d,h}} = 1412.427 e^{-1.976 \times 1.0045 \times 0.25427} \\ = 195.35$$

The Angle of incidence

$$\cos \theta = \cos(\alpha) \cos(z - z_s) \sin(\beta) + \sin(\alpha) \cos(\beta) \\ = \cos(84.38) \cos(0 - 10) \sin(30) + \sin(84.38) \cos(30) \\ = 0.91$$

The beam Radiation

$$G_{b,c} = G_{b,n} \cos \theta \\ = 818.43$$

The diffuse Radiation

$$G_{d,c} = G_{d,h} \cos^2\left(\frac{\beta}{2}\right) \\ = 195.35 \cos^2\left(\frac{30}{2}\right) \\ = 182.264 \text{ W/m}^2$$

The ground Reflection Solar Radiation

$$G_{r,c} = \rho (G_{b,n} \sin(\alpha) + G_{d,h}) \sin^2\left(\frac{\beta}{2}\right) \\ = 0.2 (898.39 \sin(84.38) + 195.35) \sin^2\left(\frac{30}{2}\right) \\ = 14.595$$

The total radiation on the tilted Collector surface

$$G_c = G_{b,c} + G_{d,c} + G_{r,c} \\ = 818.43 + 182.264 + 14.595 \\ = 1108.33 \text{ W/m}^2$$

6.4 Hourly and Daily Radiations on the Titled Surface

In order to predict the performance of a solar system, we need to calculate the hourly and daily average radiations.

① day 17 July, $n=18$, $h=-7.5$

$\therefore n=198$, From 11 to 12 AM

1) step declination angle $\delta = 23.45 \sin\left(\frac{360(284+n)}{365}\right)$

$Z_s = 10$ $= 23.45 \sin\left(\frac{360(284+198)}{365}\right)$

$\delta = 21.2$

2) step sunset hour angle $\cos(h_{ss}) = -\tan(\delta) \tan(L)$

$h_{ss} = \cos^{-1}(-\tan(26.806) \tan(21.2))$

$h_{ss} = 101.3$

3) step extraterrestrial radiation M_0

$M_0 = \frac{24 \cdot 3600}{\pi} G_{sc} \left(1 + 0.033 \cos \frac{360n}{365}\right) (\cos(L) \cos(\delta) \sin(h_{st} - h_s) \sin(L) \sin(\delta))$

$M_0 = 41.8 \text{ MJ}$

$\bar{K}_T = \frac{\bar{H}}{H_0} = \frac{29.7}{41.8} = 0.71$, H @ July 17 = 29.7 MJ/m^2

From colloration $\rightarrow \frac{\bar{H}_D}{H} = 1.390 - 4.027(\bar{K}_T) + 5.81(\bar{K}_T)^2 - 3.108(\bar{K}_T)^3$

$\bar{H}_D = 6.1366 \text{ MJ/m}^2$

② solar noon $\rightarrow h=0$

$\alpha = 0.409 + 0.5016 \sin(h_{ss} - 60)$ $\alpha' = 0.74$

$\beta = 0.669 - 0.4767 \sin(h_{ss} - 60)$ $\beta = 0.3462$

$r = \frac{\pi}{24} (\alpha + \beta \cos(h)) \frac{\cos(h) - \cos(h_{ss})}{\sin(h_{ss}) - \left(\frac{2\pi h_{ss}}{360} \cos(h_{ss})\right)} = 0.1377$

$r_d = \frac{\pi}{24} \frac{\cos(h) - \cos(h_{ss})}{\sin(h_{ss}) - \left(\frac{2\pi h_{ss}}{360} \cos(h_{ss})\right)} = 0.118$

$\bar{I} = r \cdot \bar{H} = 0.1377(29.7) = 4.089 \text{ MJ/m}^2$

$\bar{I}_d = r_d \cdot \bar{H}_D = 0.118(6.1366) = 0.7241 \text{ MJ/m}^2$

4) $\beta = 21.2$

$$R_B = \frac{\sin(L-\beta) \sin(\delta) + \cos(L-\beta) \cos(\delta) \cos(h)}{\sin(L) \sin(\delta) + \cos(L) \cos(\delta) \cos(h)}$$

$$= 0.9152$$

$$G_{BT} = G_B R_B = 1412.47 \times 0.915$$

$$= 1292.73 \text{ W/m}^2$$

to maximize it's value it should be tilt 50° @ summer
 50° @ winter

so I make it 30 to take average around the year

$$\frac{\overline{H_D}}{\overline{H}} = 1.390 - 4.027 \overline{K_T} + 5.53 \overline{K_T}^2 - 3.108 \overline{K_T}^3$$

$$\overline{H_D} = 6.1366 \text{ MJ/m}^2$$

$$h_{ss} = \min [h_{ss}, \cos^{-1}(-\tan(L-\beta) \tan(\delta))]$$

$$\cos^{-1}(-\tan((26.8206) - 30) \tan(21.21))$$

$$h_{ss} = 88.765$$

$$\overline{R_B} = \frac{\cos(L-\beta) \cos(\delta) \sin(h_{ss}) + \left(\frac{\pi}{180}\right) h_{ss} \sin(L-\beta) \sin(\delta)}{\cos(L) \cos(\delta) \sin(h_{ss}) + \left(\frac{\pi}{180}\right) h_{ss} \sin(L) \sin(\delta)}$$

$$= 0.81$$

$$\overline{R} = 0.8488 = \left(1 - \frac{\overline{H_D}}{\overline{H}}\right) \overline{R_B} + \frac{\overline{H_D}}{\overline{H}} \left(\frac{1 + \cos(\beta)}{2}\right) + 0.2 \left(\frac{1 - \cos(\beta)}{2}\right)$$

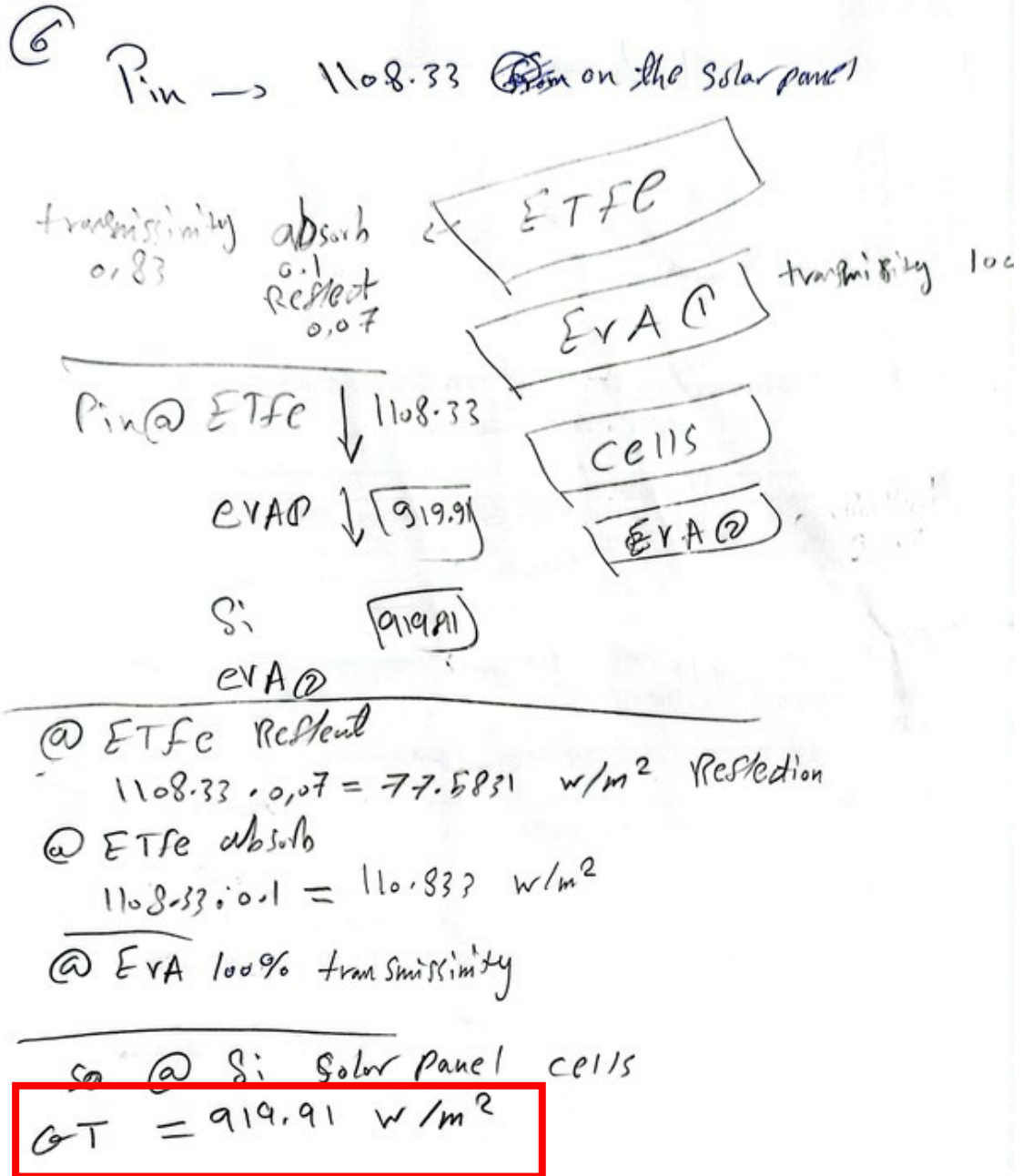
$$\overline{H_t} = \overline{R} \overline{H} = 0.8488 \times 29.7$$

$$= 25.21 \text{ MJ/m}^2 \cdot \text{day}$$

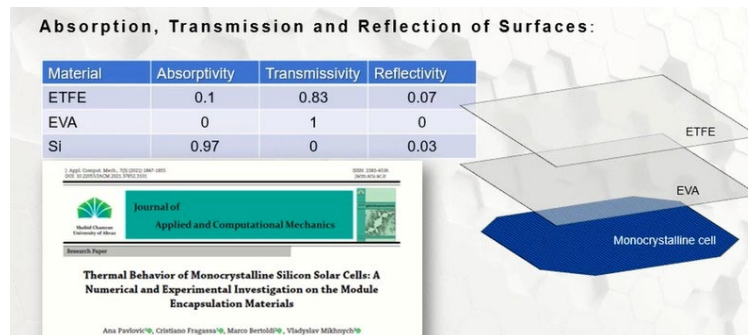
average
 daily total Radiation on
 the tilted surface July

(3)

6.5 Design and Assessing the Performance of PV



Using the calculated the original G_t (**1108.33 W/m²**) our power input, we can calculate how much actual **total radiation** is on the surface after it passes through the **absorptivity, transmittivity, and reflectivity of the panel**. We end up with a total of **919.91 w/m² total** on the surface.



Moving to the next step, we can now evaluate the efficiency and the area of the collector of the PV system. First, its important to validate our initial Power required assumption (2900 Watt), knowing the average energy consumption of a Household is around 12Kwh, the sun hours of operating, and an assumption efficiency of 0.75 (PV, generator, inverter, and battery we can calculate the power required.

5)

$$G_T = 1108.33$$

$$T = 25$$

$$\text{saturation current} = 1.7 \times 10^8 \text{ A/m}^2$$

$$\text{S.C Current denisty} = 250 \text{ A/m}^2$$

Power Req. $\rightarrow$ average Egypt house
that open (Tv, Fridge, Light, Fans, Router, water heater)
 $\rightarrow$ 12 kwh day.

$$\text{So Power Req} = \frac{\text{Daily energy need (kwh)}}{\text{Sun hours, system efficiency}}$$

$$= \frac{12}{5.5 \times 0.75} = 2.9 \text{ kw p}$$

$$= 2900 \text{ watt}$$

Our power required assumption as per calculating turned out to be correct. Moving on to maximum power input and efficiency calculation.

$$\frac{e}{kT_c} = \frac{1.602 \times 10^{-19}}{1.381 \times 10^{-23} \times (25 + 273)}$$

$$\text{voltage open circuit} = 38.927 \text{ V}^{-1}$$

$$V_{oc} = \frac{kT_c}{e} \ln\left(\frac{I_{sc}}{I_0} + 1\right) = \frac{1}{38.927} \ln\left(\frac{250}{1.7 \times 10^8} + 1\right) = 0.6014 \text{ V}$$

$$\text{To get } V_{max} \quad e^{\left(\frac{eV_{max}}{kT_c}\right)} \left(1 + \frac{eV_{max}}{kT_c}\right) = 1 + \frac{I_{sc}}{I_0}$$

$$V_{max} = 0.523 \text{ V}$$

$$I_{max} = \frac{eV_{max}}{kT_c + eV_{max}} (I_{sc} + I_0) = 238.3 \text{ A/m}^2$$

$$P_{max} = V_{max} I_{max} = 238.3 \times 0.523 = 124.63 \text{ W/m}^2$$

$$\eta = \frac{P_{max}}{P_{in}} = \frac{124.63}{919.91} = 0.13548\% = 13.548\%$$

$$A = \frac{P_{req}}{P_{max}} = \frac{2900}{125.3} = 23.14 \text{ m}^2$$

$$\text{Power generated} = 124.63 \text{ watt/m}^2$$

We notice three major outcomes:

- The maximum power input is significantly less than the power input we calculated. And that is due to losses.
- The efficiency of the system is around 13.548%, the average efficiency of PV in Egypt is around 25 ~ 28%, that is averagely good.
- The Area of the farm is 23.14 m².

6.6. Average Household power consumption

The average power consumption can be computed through calculating average daily power required from electronic device available in any household:

8) House Energy

1] AC consume 1.2 kw
(1.5 hp)

2] Refrigerator 0.2 kw (1.5-2) kwh/day

3] led light 0.01 kw (10 watt)

4] washing machine 0.5-1 kwh

5] water heater 1.5-2.5 kwh

Working hours assumption of each electronic device:

1. AC => 4 hours
2. Refrigerator => 1.8 kwh a day
3. Lights => around 5 hours a day
4. Heater=> 1 hour
5. Miscellaneous (Laptops, Tv, etc.) => 1.5 kwh a day.

* So the total Daily energy needed

① AC = 1.2 kwatt For 4 hour/day

$$E_{AC} = 1.2 \times 4 = 4.8 \text{ kwh/day}$$

② Fridge

$$E_{Fridge} = 1.8 \text{ kwh/day}$$

③ Led $\Delta = 10$

$$P = 0.01 \text{ kw}$$

usage 5 hour/day

$$E_{light} = 10 \times 0.01 \times 5 = 0.5 \text{ kwh/day}$$

④ washing machine = 0.8 kwatt

1 hour per cycle (4 times a week)

usage 0.5 cycle/day

$$E_{wm} = 0.8 \times 1 \times 0.5 = 0.4 \text{ kwh/day}$$

⑤ electric water heater

Power 2 kw

time 1 hour/day

$$E_{heater} = 2 \times 1 = 2 \text{ kwh/day}$$

CS CamSc

⑥ E_{loud} (Laptop, Tv, Router, Fans, Phones)

$$E_{loud} = 1.5 \text{ kwh/day}$$

total Daily Energy consumption

$$E_{AC} + E_{Fridge} + E_{light} + E_{washing \text{ machine}} + E_{heater} + E_{loud}$$

$$E_{total} = 4.8 + 1.8 + 0.5 + 0.4 + 2 + 1.5 = 11 \text{ kwh/day}$$

Say 12 kwh/day

Therefore, we find that the average consumption of an Egyptian household as per our calculations consumes around **12 kwh per day**, and our **PV farm** design generates around **124.63 Watts/m²** with the total area of the farm is **24 m²**.

6.6 Cost of Implementation

$$7) P = 2900 \text{ watt}$$

$$A = 23.14 \text{ m}^2$$

$$P_{\text{Generated}} = 124.63 \text{ watt/m}^2$$

The total cost For the 2.9 kw of Panels, 3KW inverter, mount
For 23m² with insallation
Estimated between 75 kw to 90kw L.E

monthly saving

11.5 kwh/day to 13 kwh/day

equall to 350 - 390 kwh / month

if the Family in Bracket 5(351-150) kwh / you Pay 1.95 L.E

monthly saving = 350 X 1.95 = 682.5 LE/month

Cost of implementations depends on the outcomes of our previous calculations, we find that the estimated cost of the **PV farm** is in-between 75k to 90k LE, using the Egyptian PV hub calculator we can input our calculations and find the cost and the actual Area of the **farm**.

(RoI) Return on investment (Pv Hub)

an usually saving = 682.5 X 12 = 8200 L.E / year

initial cost of Pv system = 90000 L.E

$$\text{Pay Back} = \frac{90000}{8200} = 10.97 \approx 11 \text{ years}$$

it will drop to 7 to 8 years once Government
implement the next planned sub cuts @ January 2026

PV Hub	Conservation
90 k	175 k
1.6 L.E/kwh	1.95 LE/kwh
10 years	14 years



Therefore, when comparing our values to the PV hub calculator we notice that our steps and outputs were within the same range so therefore our assumptions are **correct**.

7. Conclusion

By integrating real measurements, structured calculations, and diagnostic assessment, the work will provide clear insights into system efficiency, performance variability, and opportunities for engineering optimization.